

# Lecture 37 — Why did the system behave that way?

Fri Nov 20 — + Quiz 9 + Lab 5

Last updated: October 8, 2026.

Reading: [Strang §6.1](#), [Lab 5](#), and Lachniet Ch. 5 (eigenvalues/eigenvectors). Related objectives: [Lectures](#). [Schedule](#).

Same day: [Quiz 9](#) (Week 11 inverses / rank / determinants — **not** eigenvalues) and [Lab 5](#).

Bring paper. After the quiz: short lecture on software eigenpairs, dominant behavior, and a normal-mode demo, then Lab 5.

## Learning objectives

By the end of class, students should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify  $Av \approx \lambda v$  numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

## Warm-up

Let

$$A = \begin{bmatrix} 1.1 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Hand-compute  $Av$ . Guess a plausible  $\lambda$  so that  $Av \approx \lambda v$ . (Exact eigenpairs come from software in lab.)

## Fill in

**Software eigenpairs.** Octave `eig(A)` / Python `np.linalg.eig(A)` return eigenvalues and a matrix whose columns are eigenvectors.

**Verify.** For each reported pair  $(\lambda, v)$ , check

$$Av \approx \underline{\hspace{2cm}}$$

(componentwise; floating-point noise is expected). Never trust a bare printout without this check.

**Dominant behavior.** If  $|\lambda_1| > |\lambda_2| \geq \dots$  and  $x_0$  has a component along the  $\lambda_1$ -eigenvector, then for large  $t$

$$A^t x_0 \approx c \lambda_1^t v_1.$$

That explains Lab 4-style growth/decay/stabilization once you know the eigenpairs.

**Complex eigenvalues.** For a real matrix, nonreal eigenvalues come in conjugate pairs. In the real plane they produce \_\_\_\_\_ (rotation / oscillation), matching Monday's pure-rotation example.

**Normal modes (demo, not a DE course).** For a symmetric mass–spring style matrix, eigenvectors are often \_\_\_\_\_ (orthogonal). Software shows the modes; we do **not** derive differential equations today.

## Worked in class

**W1.** Software sketch. On the board / laptop, for  $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ :

```
# Octave: [V,D] = eig(A)      or      eig(A)
# Python: vals, vecs = np.linalg.eig(A)
```

Read off  $\lambda$ 's and columns of  $V$ . Verify  $Av_1 = \lambda_1 v_1$  for one column.

**W2.** Explain a simulation. Same  $A$  as W1;  $x_0 = (1, 0)$ . Using eigenpairs, explain in one sentence why  $x_t$  grows and which direction it approaches.

**W3.** Complex / oscillation. Software on  $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$  should report eigenvalues near  $\pm i$ . Connect that to the cycling you saw in Lab 4 stretch / Lecture 35.

**W4.** Symmetric demo. Let  $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  (symmetric). Check that the two eigenvectors from W1 are orthogonal ( $v_1^\top v_2 = 0$ ). Normal-mode pictures use the same idea in higher dimension.

## Your turn

**Y1.** Pseudocode: given  $A$ , obtain eigenpairs in software and write the one-line check that  $Av \approx \lambda v$  for the first pair.

**Y2.** True or false: if  $|\lambda_1| = 1.5$  and  $|\lambda_2| = 0.5$ , then for generic  $x_0$  the long-run direction of  $A^t x_0$  is the  $\lambda_1$ -eigendirection.

**Y3.** Quiz / lab reminder. In one sentence: what does Quiz 9 cover today, and what does Lab 5 cover?

## Lab 5, same day

After the short lecture: [Lab 5 — Why Did the System Behave That Way?](#). Compute eigenpairs with software, verify  $Av \approx \lambda v$ , explain the Lab 4 mystery, interpret complex eigenvalues, and peek at a symmetric / normal-mode example. Preview the [one-pager](#); write on the hard copy in class.

**Reminder:** [Quiz 9](#) today covers Week 11 (inverses / rank / determinants), not this eigen lab.

## Answers (Your turn)

Y1. Sample: `vals, V = np.linalg.eig(A)` then `A @ V[:,0] - vals[0]*V[:,0]` (should be near 0); Octave: `[V,D]=eig(A)` then `A*V(:,1) - D(1,1)*V(:,1)`.

Y2. **True.** The  $\lambda_1$  term grows relative to  $\lambda_2$ , so the state aligns with  $v_1$  (unless the  $v_1$ -component of  $x_0$  is exactly zero).

Y3. Quiz 9 is Week 11 (inverses / rank / determinants); Lab 5 is software eigenpairs and explaining dynamics — not on the quiz.